3.1.7 Allotropic forms of phosphorus:

Phosphorus has several allotropic modification of which the three forms namely white, red and black phosphorus are most common.

The freshly prepared white phosphorus is colourless but becomes pale yellow due to formation of a layer of red phosphorus upon standing. Hence it is also known as yellow phosphorus. It is poisonous in nature and has a characteristic garlic smell. It glows in the dark due to oxidation which is called phosphorescence. Its ignition temperature is very low and hence it undergoes spontaneous combustion in air at room temperature to give P_2O_5.

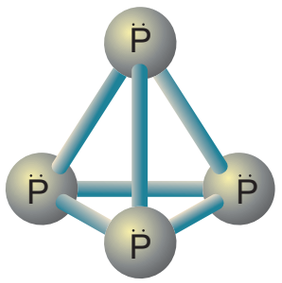


Figure 3.2 Structure of white phosphorus

The white phosphorus can be changed into red phosphorus by heating it to 420 ^{0}C in the absence of air and light. Unlike white phosphorus it is not poisonous and does not show Phosphorescence. It also does not ignite at low temperatures. The red phosphorus can be converted back into white phosphorus by boiling it in an inert atmosphere and condensing the vapour under water.

The phosphorus has a layer structure and also acts as a semiconductor. The four atoms in phosphorus have polymeric structure with chains of P_4 linked tetrahedrally. Unlike nitrogen P≡P is less stable than P-P single bonds. Hence, phosphorus atoms are linked through single bonds rather than triple bonds. In addition to the above two more allotropes namely scarlet and violet phosphorus are also known for phosphorus.

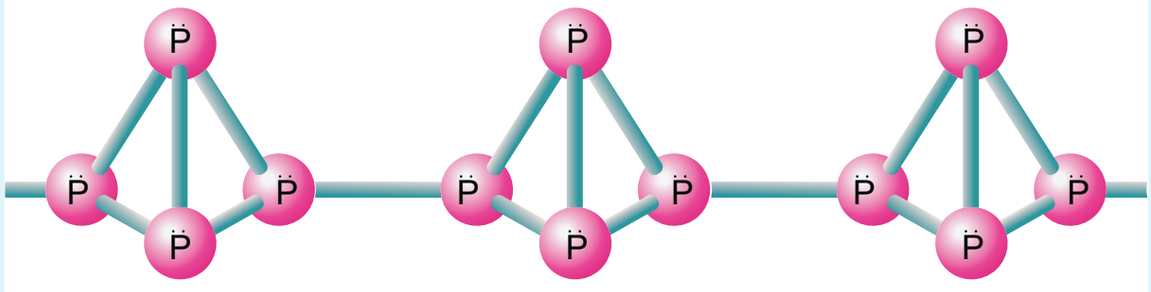


Figure 3.3 Structure of red phosphorus

3.1.8 Properties of phosphorus

Phosphorus is highly reactive and has the following important chemical properties

Reaction with oxygen: Yellow phosphorus readily catches fire in air giving dense white fumes of phosphorus pentoxide. Red phosphorus also reacts with oxygen on heating to give phosphorus trioxide or phosphorus pentoxide.

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